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Assignment # 03

Subject:- Artificial Intelligence (CS-860)

Topic:- Constraint satisfaction problem.

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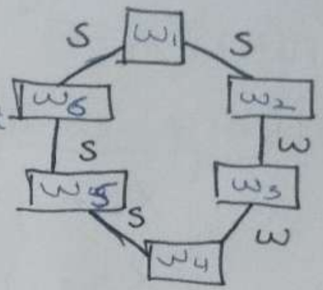
Date of Submission:
02 - March - 2026.

(a) State the binary and unary constraints for this (1) CSP (either implicit or explicit)?

Strong (S)	one of two corridors is a pit
weak (W)	at least one is exit, Neither pit
No breeze	Both are ghost.



Unary constraints

(1) $X_2 - X_3 = W \rightarrow X_2 \neq P$ and $X_3 \neq P$.

(2) $X_3 - X_4 = W \rightarrow X_3 \neq P$ and $X_4 \neq P$.

Domains after unary constraints:-

$X_1 = (P, G, E)$, $X_4 = (G, E)$

$X_2 = (G, E)$, $X_5 = (P, G, E)$

$X_3 = (G, E)$, $X_6 = (P, G, E)$

Binary constraints:-

- $X_6 = P$ or $X_1 = P$
- $X_1 = P$ or $X_2 = P$
- $X_4 = P$ or $X_5 = P$
- $X_5 = P$ or $X_6 = P$

From S breeze at least one must P.

- $X_2 = E$ or $X_3 = E$
- $X_3 = E$ or $X_4 = E$

From W-breeze → at least one must E.

- Not ($X_i = E$ and $X_j = E$) → for all 6 adjacent pairs.

(b) Suppose we assign X_1 to E. perform foward che[ck]
after this assignment. Also enforce unary constraints.

After $X_1 = E$

(a) Unary constraints: X_2, X_3, X_4 lose P.

$X_2 = \{G, E\}$.
$X_3 = \{G, E\}$.
$X_4 = \{G, E\}$.

(b) forward checking.

⊙ $X_1 - X_2 = S$, $X_1 \neq P$, $\rightarrow X_2$ must be P.

But $X_2 = \{G, E\} \longrightarrow$ therfore
$X_2 = \{\ \}$ (empty)

⊙ $X_6 - X_1 = S$: $X_1 \neq P \rightarrow X_6$ must be P.

Domain wipe out at X_2 therefore concolusion is that $X_1 = E$ is INVALID.

(c) Suppose forward checking returns the following set of possible assignments

according to MRV, which variable or variables could the solver assign first?

X_1	P		
X_2		G	E
X_3		G	E
X_4		G	E
X_5	P		
X_6	P	G	E

MRV selects variable with minimum remaining values.

$X_1 = \{P\} \longleftarrow$ MRV choice
$X_5 = \{P\} \longleftarrow$ MRV choice
$X_2 = \{G, E\}$
$X_3 = \{G, E\}$
$X_4 = \{G, E\}$
$X_6 = \{P, G, E\}$

therefore X_1 & X_5
(both have $\underset{\text{min}}{\downarrow}$ value = 1).

(d) Assume that pacman knows that $X_6 = G$. list all the solutions of this CSP or write non if no solution exists.

Given $X_6 = G$.

<u>Forced</u> $X_1 = P$ $(X_1 - X_2 = S\ ,\ X_2 \neq P)$

$X_5 = P$ $(X_4 - X_5 = S\ ,\ X_4 \neq P)$

Test = $X_2, X_3, X_4 = \{G, E\}$.

valid solutions

a $\{X_1 = P, X_2 = G, X_3 = E, X_4 = G, X_5 = P, X_6 = G\}$

b $\{X_1 = P, X_2 = E, X_3 = G, X_4 = E, X_5 = P, X_6 = G\}$

→ 2 solutions exists.

Q NO 2.

(a) Formulate this problem as CSP problem in which there is one variable per class starting the domains (after enforcing unary constraints) and binary constraints constrainst should be specified formally and precisely. but may be implicit rathar than explicit.

<u>variables are</u> :- C_1, C_2, C_3, C_4, C_5

<u>Domains after unary constraints</u>.

$C_1 = \{A, C\}$, $C_4 = \{B, C\}$.

$C_2 = \{A\}$. , $C_5 = \{A, B\}$.

$C_3 = \{B, C\}$

Binary constraints:-

1. $C_1 \neq C_2$ overlap $C1 \rightarrow C_2$
2. $C_2 \neq C_3$ overlap $C_2 \rightarrow C_3$
3. $C_2 \neq C_4$ overlap $C_2 \rightarrow C_4$
4. $C_3 \neq C_4$ overlap $C_3 \rightarrow C_4$

(b) Draw the constraint graph associated with your CSP.